

- **Example1:** IP address: 10.1.8.200/26

Subnet mask: 255.255.255.192 N.N.N.nnnhhhhh

Binary for IP address of each octet:

00001010. 00000001. 00001000. 11001000

Binary for subnet mask of each octet:

11111111. 11111111. 11111111. 11000000

Use the AND logic to get the network address

00001010. 00000001. 00001000. 11000000

Hence the network address in decimal

10.1.8.192

Broadcast address: converting the h bits into 1s: 10.1.8.255

First host: 10.1.8.193 **last host:** 10.1.8.254

- **Example2:** IP address: 10.1.1.101/25

Subnet mask: 255.255.255.128 N.N.N.nhhhhhhh

Binary for IP address of each octet:

00001010. 00000001. 00000001. 01100101

Binary for subnet mask of each octet:

11111111. 11111111. 11111111. 10000000

Use the AND logic to get the network address

00001010. 00000001. 00000001. 00000000

Hence the network address in decimal

10.1.1.0

Broadcast address: converting the h bits into 1s

Note: here the last octet of the broadcast isn't 255 as the last octet of the network address is all 0s (00000000) and this has one n and seven h. We only convert the h into 1s so the broadcast address here: **Broadcast address:** 10.1.1.127

First host: 10.1.1.1 **last host:** 10.1.1.126

- **Example3:** IP address: 209.165.200.227/27

Subnet mask: 255.255.255.224 N.N.N.nnnhhhhh

Binary for IP address of each octet:

11010001. 10100101. 11001000. 11100011

Binary for subnet mask of each octet:

11111111. 11111111. 11111111. 11100000

Use the AND logic to get the network address

11010001. 10100101. 11001000. 11100000

Hence the network address in decimal

209.165.200.224

Broadcast address: converting the h bits into to 1s: 209.165.200.255

First host: 209.165.200.225 **last host:** 209.165.200.254

- **Example4:** IP address: 172.16.117.77/20

Subnet mask: 255.255.240.0 N.N.nnnnhhhh.H

Binary for IP address of each octet:

10101100. 00010000. 01110101. 01001101

Binary for subnet mask of each octet:

11111111. 11111111. 11110000. 00000000

Use the AND logic to get the network address

10101100. 00010000. 01110000. 00000000

Hence the network address in decimal

172.16.112.0

Broadcast address: converting the h bits into to 1s (here we have four h from the third octet and H for the fourth octet so we should have 12 1s for the broadcast): 172.16.127.255

First host: 172.16.112.1 **last host:** 172.16.127.254

Note: first host is network address +1 while last host is broadcast address -1